

Forensic Acquisition and Analysis of a storage device using Guymager and Autopsy - Documentation

Specification of the device which the task is done in:

Processor	Intel core i5 10th gen (10300H)
RAM	16 GB
Storage	512 GB NVME SSD
Virtualization	Oracle VirtualBox
Guest OS	Ubuntu 22

VirtualBox Specifications:

Base memory	11329MB
Video Memory	16MB
Storage Allocated	50GB

Installing Guymager on Ubuntu 22 by using the following commands :

```
sudo apt update && sudo apt upgrade -y  
sudo apt install guymager -y
```

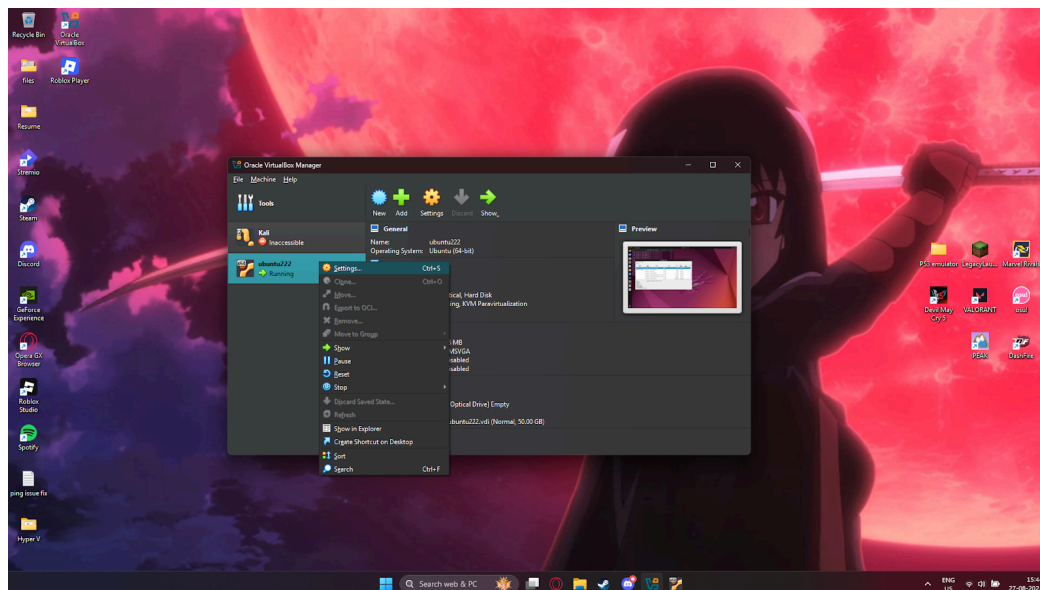
To verify installation:

Go to Terminal and type Guymager, It should open up.

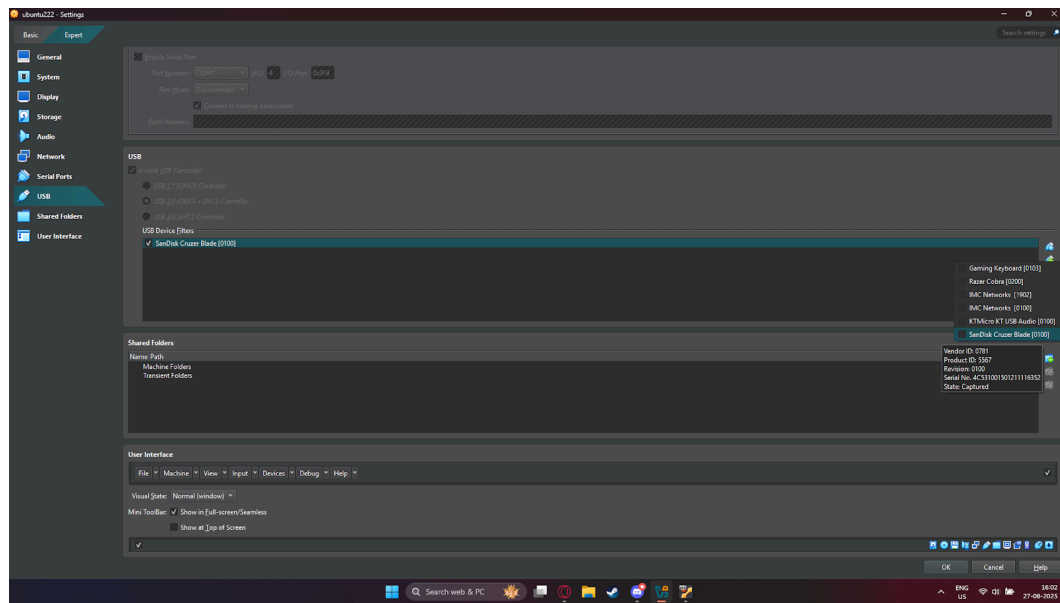
Acquiring the disk image of a storage device using Guymager:

Since this is done on a Virtual Machine, we should add the storage device (which in this case, i am using a pen drive) manually in the Virtual Machine settings.

To do that, Go to the Ubuntu 22 Virtual Machine settings in Oracle VirtualBox.



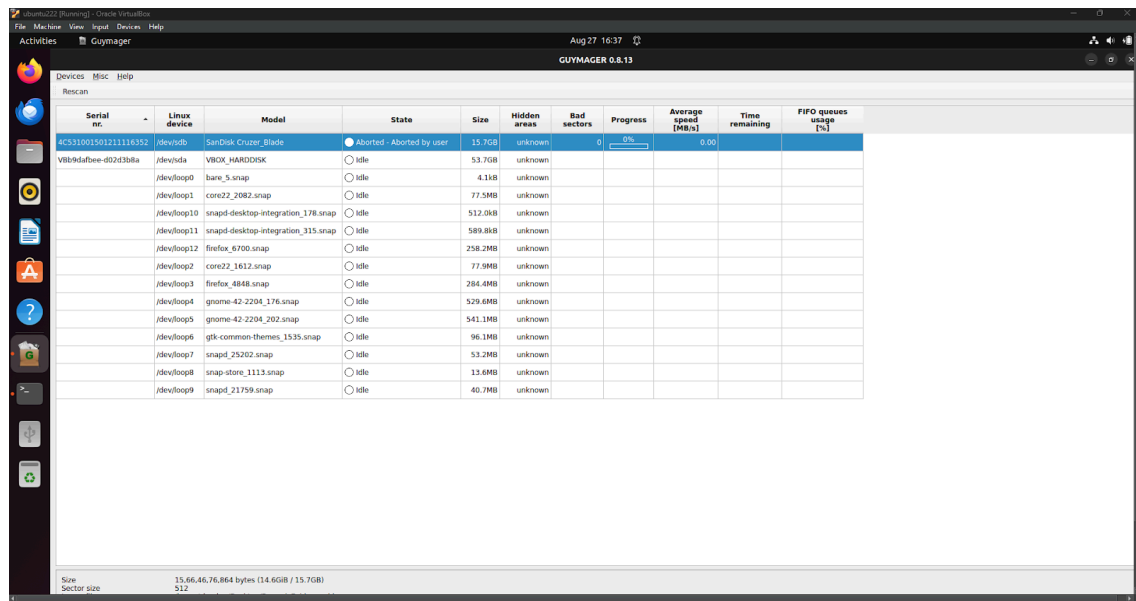
Locate the USB option in the settings menu and then add the storage device you want it to appear inside the ubuntu 22 VM.



Now open the Ubuntu 22 VM and open the guymager application with root access using the command:

```
sudo guymager
```

After opening GuyMager, Now you should see the storage device listed in there like this,



Now right click the storage device and then select the option Acquire Image.

You will be prompted with a window like this ,

Acquire image of /dev/sdb

File format

☐ Linux dd raw image (file extension .dd or .xxx)
 ☒ Split image files

☒ Expert Witness Format, sub-format Guymager (file extension .Exx)
 Split size

Case number

Evidence number

Examiner

Description

Notes

Destination

Image directory

Image filename (without extension)

Info filename (without extension)

Hash calculation / verification

☒ Calculate MD5
 ☐ Calculate SHA-1
 ☐ Calculate SHA-256

☐ Re-read source after acquisition for verification (takes twice as long)

☒ Verify image after acquisition (takes twice as long)

There are two formats which you can acquire the disk image with,

.dd format: Bit for bit copy of the source drive

.E01 format : Proprietary forensic image format.

You can choose any format according to your requirements.

After filling all the fields, you can start the process. After the process is done, you can locate the file on which path you gave on the image directory field.

Autopsy Installation:

Open terminal on ubuntu VM and type these commands:

```
sudo apt update
```

```
sudo apt install snapd
```

```
sudo apt snap install autopsy
```

To verify its installation, type autopsy in the terminal , the tool must open up.

Investigating a disk image in Autopsy

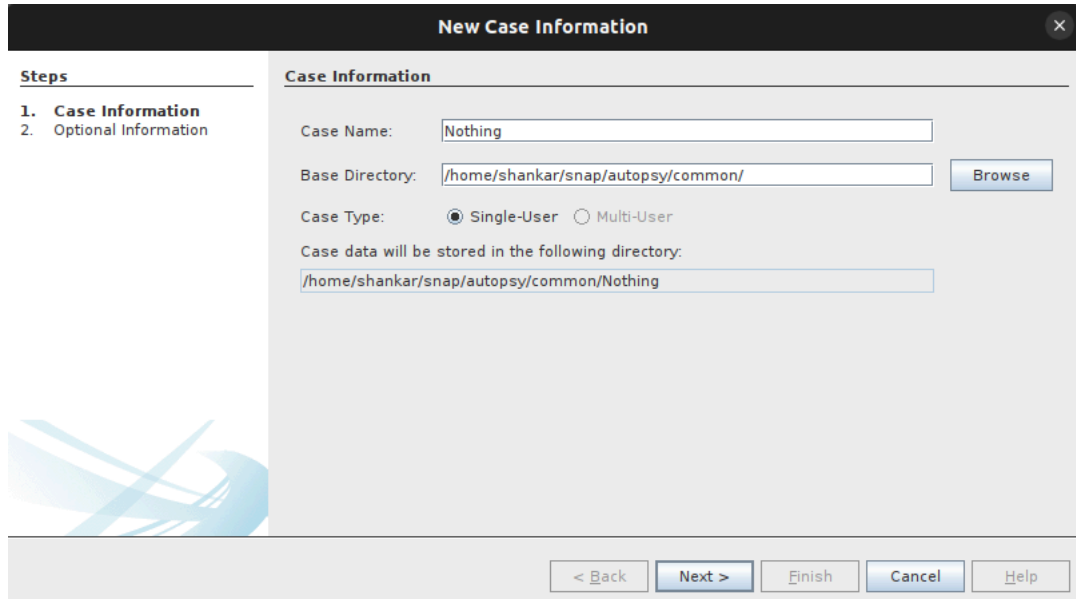
For this investigation I acquired this sample image disk from

<https://dftt.sourceforge.net>

When opening autopsy, you will greeted with window like this,



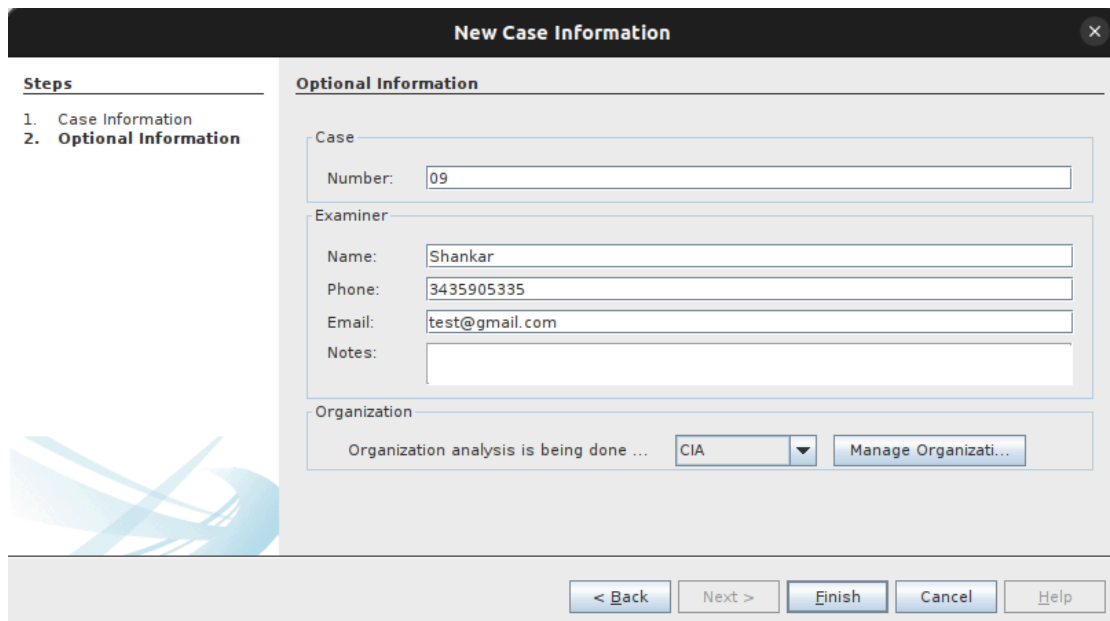
Open a new case.



The dialog box is titled "New Case Information" and has a close button (X) in the top right corner. On the left, a "Steps" panel shows "1. Case Information" as the current step and "2. Optional Information" as the next step. The main area is titled "Case Information" and contains the following fields:

- Case Name:** A text box containing "Nothing".
- Base Directory:** A text box containing "/home/shankar/snap/autopsy/common/" with a "Browse" button to its right.
- Case Type:** Two radio buttons: "Single-User" (selected) and "Multi-User".
- Case data will be stored in the following directory:** A text box containing "/home/shankar/snap/autopsy/common/Nothing".

At the bottom, there are five buttons: "< Back", "Next >", "Finish", "Cancel", and "Help".

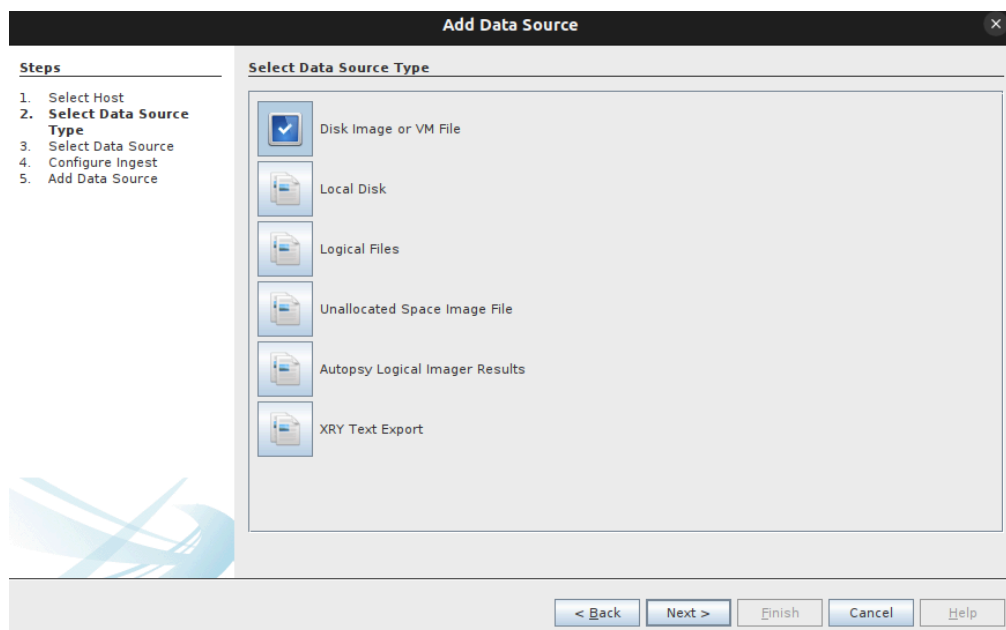
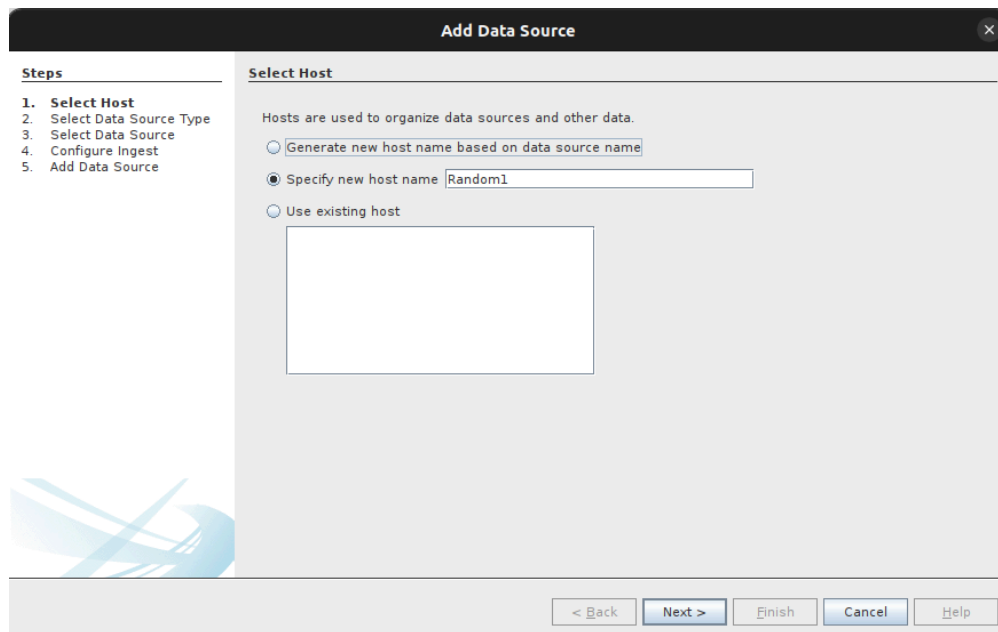


The dialog box is titled "New Case Information" and has a close button (X) in the top right corner. On the left, a "Steps" panel shows "1. Case Information" and "2. Optional Information" as the current step. The main area is titled "Optional Information" and contains the following fields:

- Case:** A text box containing "09".
- Examiner:** A group box containing four text boxes: "Name" (Shankar), "Phone" (3435905335), "Email" (test@gmail.com), and "Notes" (empty).
- Organization:** A group box containing a text box with "Organization analysis is being done ...", a dropdown menu showing "CIA", and a "Manage Organizati..." button.

At the bottom, there are five buttons: "< Back", "Next >", "Finish", "Cancel", and "Help".

After this, add a new data source.



Since we are going to analyse disk image choose disk image

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
- 3. Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Path:

☐ Ignore orphan files in FAT file systems

Time zone:

Sector size:

Bitlocker Password (optional):

Hash Values (optional):

MDS:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

After this, configure your ingest based on your investigation requirements.

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. Select Data Source
- 4. Configure Ingest**
5. Add Data Source

Configure Ingest

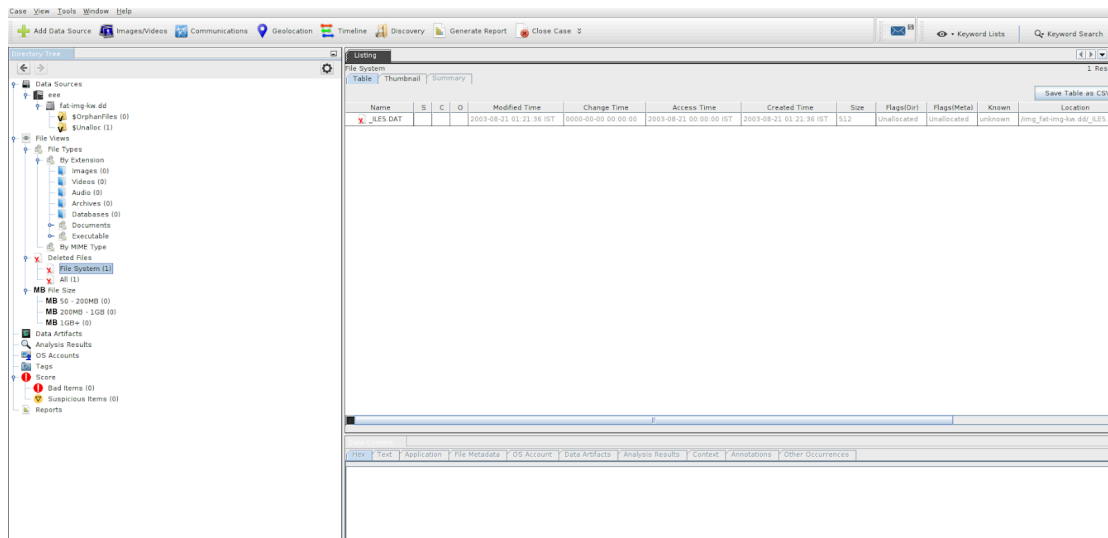
Run ingest modules on:

- ☒ Encryption Detection
- ☒ Interesting Files Identifier
- ☒ Central Repository
- ☒ PhotoRec Carver
- ☒ Virtual Machine Extractor
- ☒ Data Source Integrity
- ☒ Android Analyzer (aLEAPP)
- ☐ Cyber Triage Malware Scanner
- ☒ DJI Drone Analyzer
- ☐ Plaso
- ☒ YARA Analyzer
- ☒ iOS Analyzer (iLEAPP)
- ☒ GPX Parser
- ☒ Android Analyzer

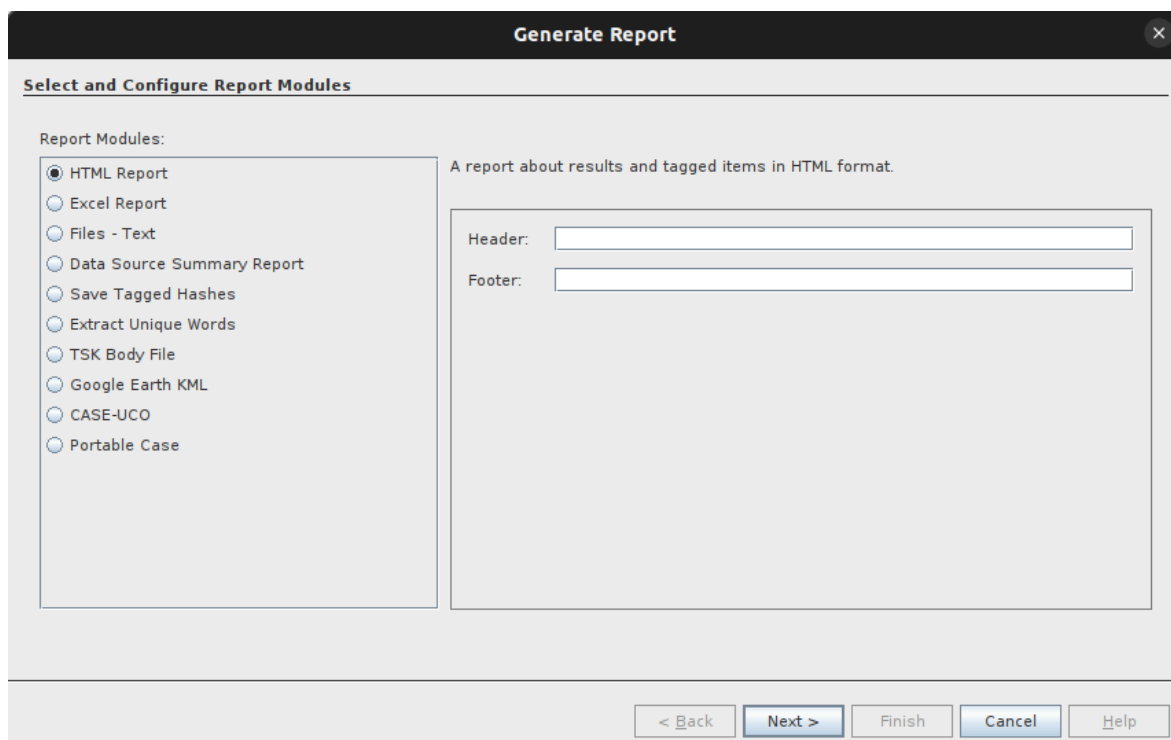
The selected module has no per-run settings.

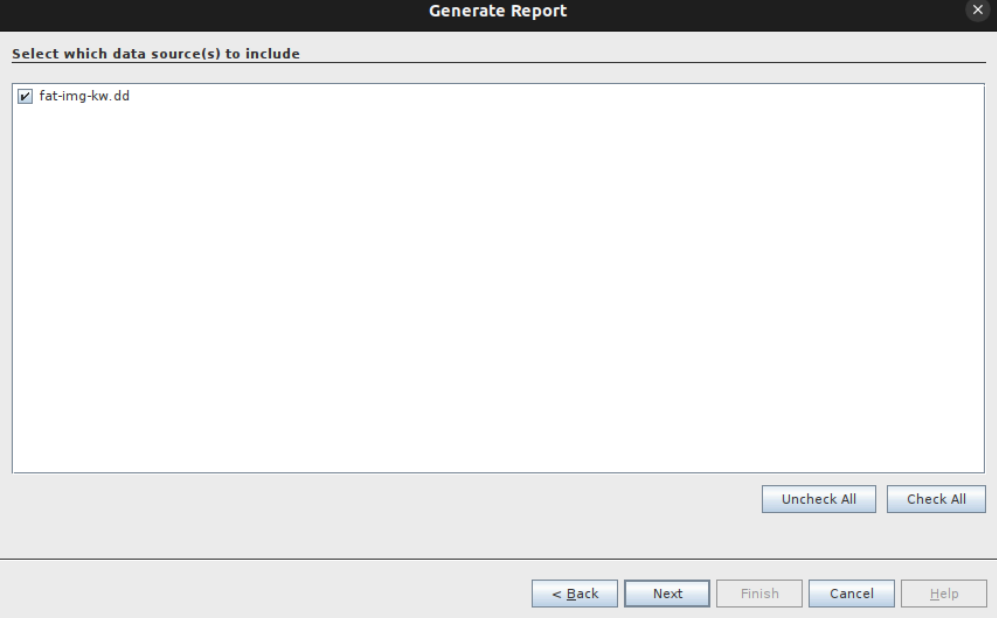
Extracts recent user activity, such as Web bro...

After processing you will be greeted with this, where it shows the directories in the data source



After analysing it, create a report by clicking the generate report option. You can choose the format of the report like this which is shown below,





The image shows a 'Generate Report' dialog box. It has a title bar with the text 'Generate Report' and a close button (X). Below the title bar is a section titled 'Select which data source(s) to include'. This section contains a list box with one item, 'fat-img-kw.dd', which is checked. At the bottom right of the list box area are two buttons: 'Uncheck All' and 'Check All'. At the very bottom of the dialog box are five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Generate Report

Select which data source(s) to include

☒ fat-img-kw.dd

Uncheck All Check All

< Back Next Finish Cancel Help

After this your report is created successfully.